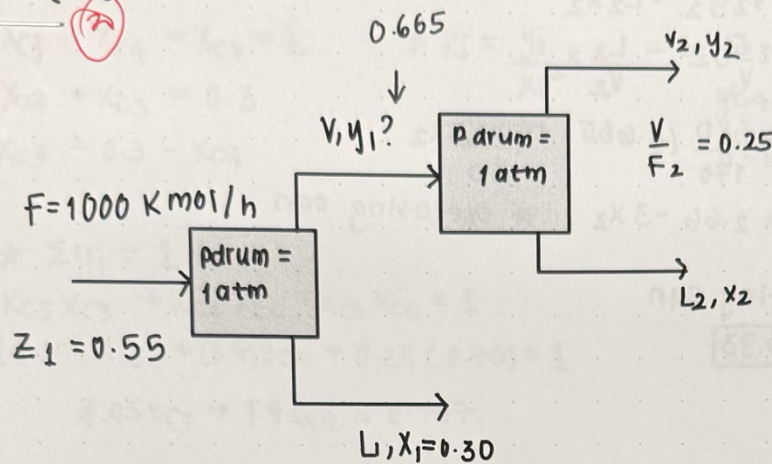


VIANCA ESCALANTE

1. (2D5)



i.) First drum

★ Table 2.8

$$x_1 = 0.3 \quad y_1 = 0.665 \quad T_1 = 78^\circ\text{C}$$

★ overall m.b: $F = V_1 + L_1 \Rightarrow L_1 = F - V_1$

comp m.b: $Fz_1 = V_1y_1 + L_1x_1$

$$Fz_1 = V_1y_1 + (F - V_1)x_1$$

$$Fz_1 = V_1y_1 + Fx_1 - V_1x_1$$

$$Fz_1 - Fx_1 = V_1y_1 - V_1x_1$$

$$F(z_1 - x_1) = V_1(y_1 - x_1)$$

$$z_1 - x_1 = \frac{V_1}{F} (y_1 - x_1)$$

$$\frac{V_1}{F} = \frac{(z_1 - x_1)}{(y_1 - x_1)}$$

$$\frac{V}{F} = \frac{(0.55 - 0.3)}{(0.665 - 0.3)}$$

$$\frac{V}{F} = 0.68$$

$$V_1 = 0.68(1000) = 680 \frac{\text{kmol}}{\text{hr}}$$

$$0.6875$$

$$687.5 \frac{\text{kmol}}{\text{hr}}$$

ii.) 2nd drum

★ $F = V_1$ from first drum

$$z = y_1$$

$$\frac{V_2}{F} = 0.25$$

$$V_2 = (0.25)(680) = 170 \frac{\text{kmol}}{\text{hr}}$$

$$171.875 \frac{\text{kmol}}{\text{hr}}$$

★ overall m.b: $F = V_2 + L_2 \Rightarrow L_2 = 680 - 170 = 510 \frac{\text{kmol}}{\text{hr}}$

★ component m.b: $Fz = V_2 y_2 + L_2 x_2$

$$y_2 = \frac{F}{V_2} z - \frac{L_2}{V_2} x_2$$

$$y_2 = \frac{680}{170} (0.665) - \frac{510}{170} x_2$$

$$y_2 = 2.66 - 3x_2 \quad \star \text{ operating eqn}$$

★ graph VLE data & operating eqn

$$x_2 = 0.61$$

$$y_2 = 0.83$$

★ interpolate VLE data

$$x_2: T = T_a + \frac{x - x_a}{x_b - x_a} (T_b - T_a) = 71.2 + \frac{0.61 - 0.6}{0.7 - 0.6} (69.3 - 71.2)$$

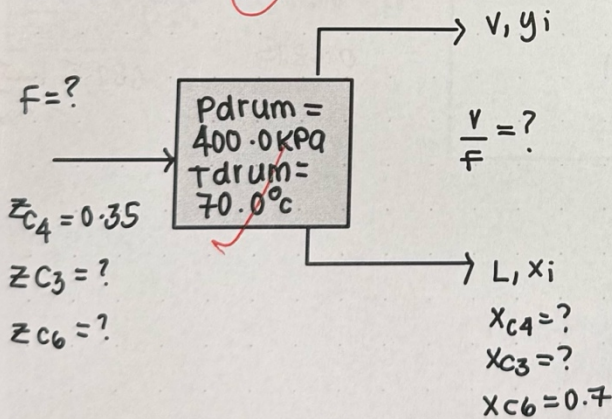
$$T = 71.01^\circ\text{C}$$

$$y_2: T = 71.2 + \frac{0.83 - 0.825}{0.87 - 0.825} (69.3 - 71.2)$$

$$T = 70.99^\circ\text{C}$$

$$T_2 = \frac{71.01 + 70.99}{2} = 71^\circ\text{C}$$

2. (2.D10)

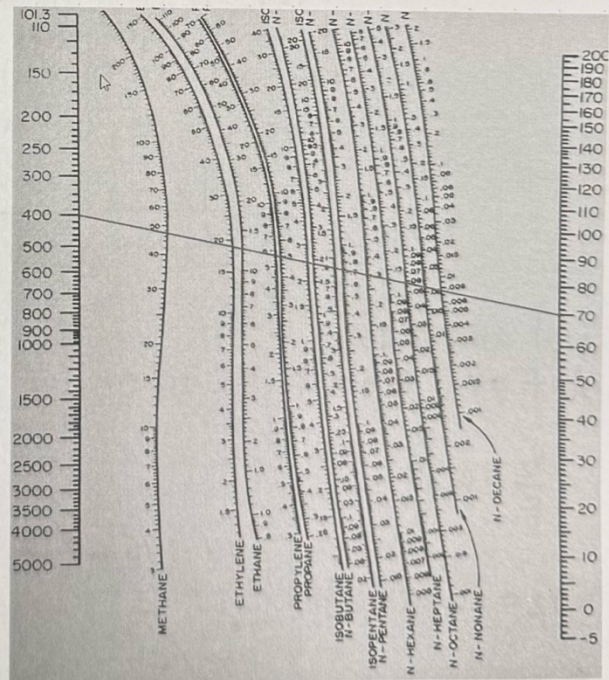


★ K values

$$K_{C3} = 4.95$$

$$K_{C4} = 1.9$$

$$K_{C6} = 0.29$$



$$x_{c4} + x_{c6} = 1$$

$$x_{c4} + x_{c3} = 0.3$$

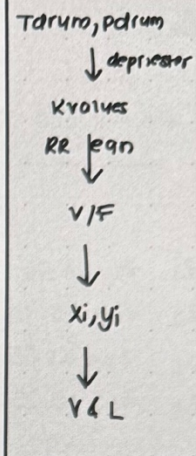
$$x_{c3} = 0.3 - x_{c4}$$

$$\star K_i = \frac{y_i}{x_i} \Rightarrow$$

$$y_{c3} = K_{c3} x_{c6}$$

$$y_{c4} = K_{c4} x_{c4}$$

$$y_{c6} = K_{c6} x_{c6}$$



$$\star \sum y_i = 1$$

$$K_{c3} x_{c3} + K_{c4} x_{c4} + K_{c6} x_{c6} = 1$$

$$(4.95) x_{c3} + (1.9) x_{c4} + 0.29 (0.76) = 1$$

$$4.95 x_{c3} + 1.9 x_{c4} = 0.797$$

$$\star \text{plug } x_{c3}$$

$$4.95 (0.3 - x_{c4}) + 1.9 x_{c4} = 0.797$$

$$1.485 - 4.95 x_{c4} + 1.9 x_{c4} = 0.797$$

$$-3.05 x_{c4} = -0.688$$

$$x_{c4} = 0.226$$

⇒

$$\star y_i = K_i x_i$$

$$y_{c3} = 4.95 (0.074) = 0.3663$$

$$y_{c4} = 1.9 (0.226) = 0.4294$$

$$y_{c6} = 0.29 (0.7) = 0.203$$

$$x_{c3} = 0.3 - 0.226 = 0.074 = x_{c3}$$

$$\star \text{m.B: } F = Z + L \rightarrow L = F - V$$

$$\text{comp.B: } f z = v y + L x$$

$$z = \frac{v}{F} y + \frac{L}{F} x$$

$$z = \frac{v}{F} y + \frac{F-v}{F} x$$

$$z = \frac{v}{F} y + (1 - \frac{v}{F}) x$$

$$\star \text{plug } z_{c4} = 0.35$$

$$0.35 = \frac{v}{F} (0.4294) + (1 - \frac{v}{F}) 0.226$$

$$0.35 = 0.4294 \frac{v}{F} + 0.226 - 0.226 \frac{v}{F}$$

$$0.124 = 0.2034 \frac{v}{F}$$

$$\frac{v}{F} = 0.61 \quad \text{0.55}$$

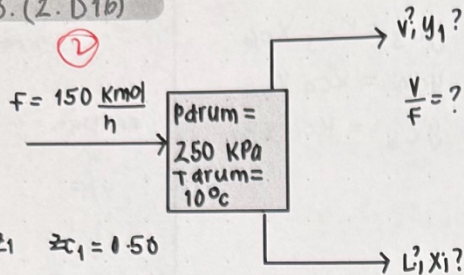
$$\star \text{Find } z_{c6}$$

$$z_{c6} = 0.61 (0.203) + (1 - 0.61) 0.7$$

$$= 0.12383 + 0.273$$

$$z_{c6} = 0.397 \quad \approx 0.4$$

3. (2. D1b)



$z_1 = z_{c1} = 0.50$
 $z_2 = z_{c4} = 0.10$
 $z_3 = z_{c5} = 0.15$
 $z_4 = z_{c6} = 0.25$

★ K values

$K_{c1} = 51$

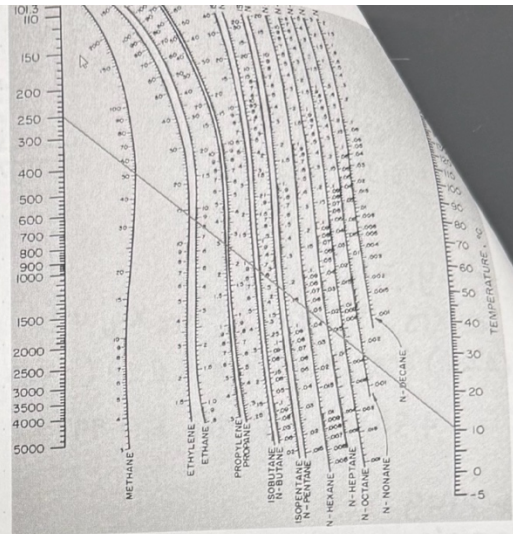
$K_{c4} = 0.62$

$K_{c5} = 0.17$

$K_{c6} = 0.05$

★ RR

$$\sum \frac{(K_i - 1)z_i}{1 + (K_i - 1)V/F} = 0$$



★ Use excel to do iterations import z_i, K_i , and RR equation start with a guess $0 \leq \frac{V}{F} \leq 1$, goal seek RR equation to 0 by changing V/F where it equals 0.58

$\frac{V}{F} = 0.58$

$V = 0.58(150) = 87 \text{ kmol/hr}$

$L = F - V = 150 - 87 = 63 \text{ kmol/hr}$

★ $x_i = \frac{z_i}{1 + (K_i - 1)V/F}$ $y_i = K_i x_i$

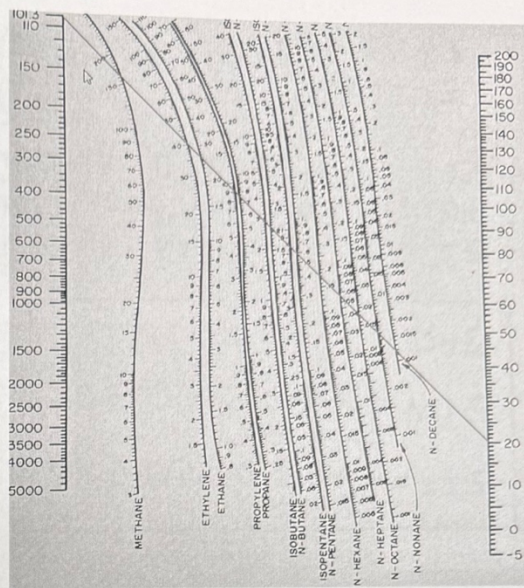
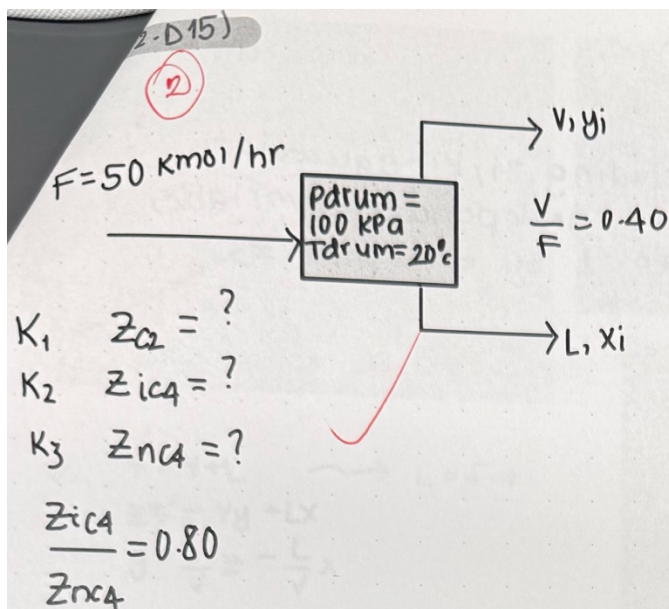
$x_{c1} = \frac{0.5}{1 + (51 - 1)0.58} = 0.017$

$y_1 = 51(0.017) = 0.867$

↓ repeat rest

$x_{c2} = 0.13$ $y_{c2} = 0.079$
 $x_{c3} = 0.29$ $y_{c3} = 0.049$
 $x_{c4} = 0.56$ $y_{c4} = 0.028$

Component	z_i	K_i	$K_i - 1$	RR
C1	0.5	51	50	0.825977
C4	0.1	0.62	-0.38	-0.04887
C5	0.15	0.17	-0.83	-0.24214
C6	0.25	0.05	-0.95	-0.535
			sum	-3.6E-05
V/F	0.585344			



- 1.) Use dePriester chart to find values of the 3 components by creating a diagonal line from 100 kPa on the left to 20°C on the right.

$K_1 = 27.2$

★ double check using eq 2.28

$K_2 = 2.92$

$K_3 = 2.16$

- 2.) write down equations needed, in terms of z_3

$1 = z_1 + z_2 + z_3$ $\frac{z_2}{z_3} = 0.80 \Rightarrow z_2 = 0.80 z_3$

$1 = z_1 + 0.80 z_3 + z_3 \Rightarrow z_1 + 1.8 z_3 = 1 \Rightarrow z_1 = 1 - 1.8 z_3$

RR: $\sum \frac{(K_i - 1)z_i}{1 + (K_i - 1)V/F} = 0 \Rightarrow$ can't be used multiple unknowns so... $\Rightarrow \sum \frac{K_i z_i}{1 + (K_i - 1)V/F} = 1$
vapor

- 3.) plug in for each of the components

1: $1 + 0.40(27.2 - 1)$
 $= 1 + 0.40(26.2)$
 $= 11.48$

2: $1 + 0.40(2.92 - 1)$
 $= 1 + 0.40(1.92)$
 $= 1.768$

3: $1 + 0.40(2.16 - 1)$
 $= 1 + 0.40(1.16)$
 $= 1.464$

$\frac{27.2 z_1}{11.48} + \frac{2.92 z_2}{1.768} + \frac{2.16 z_3}{1.464} = 1$

$$\left. \begin{aligned} z_1 &= 1 - 1.80 z_3 \\ z_3 &= 0.80 z_3 \end{aligned} \right\} \text{plug in } \uparrow$$

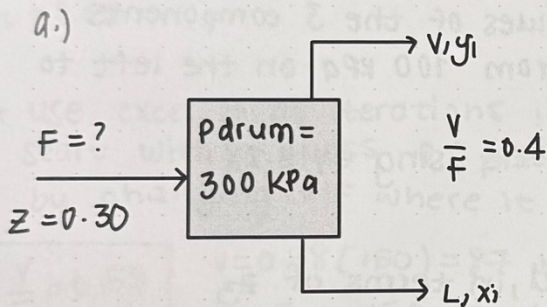
4.) Set up in excel a table including z_i, K_i , equation have an initial guess for our independent variable, z_3 and goalseek KR can to 1 by changing z_3

$$z_{nc4} = 0.30$$

$$z_{ic4} = 0.80 (0.30) = 0.24$$

$$z_{c2} = 1 - 1.8(0.30) = 0.46$$

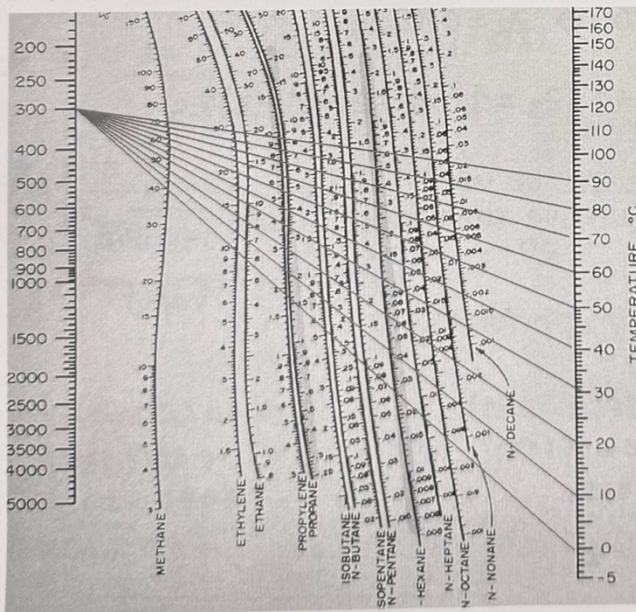
5. (2-D24)



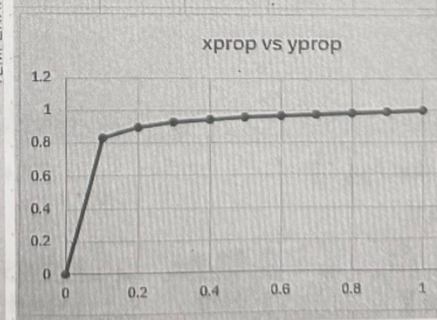
$$y_i = K_i x_i$$

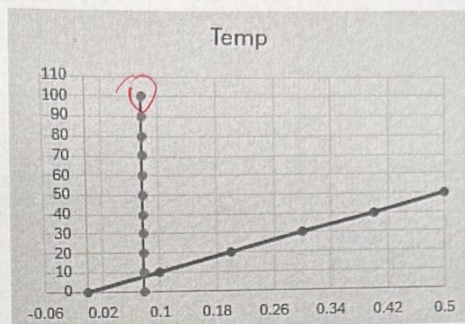
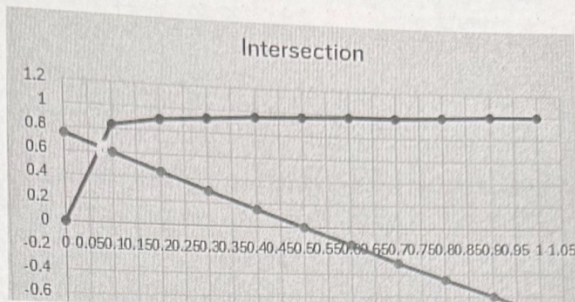
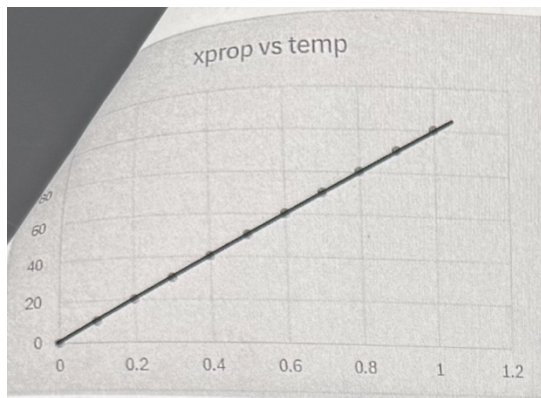
$$\sum y_i = 1$$

$$y = \frac{K_1 x_1}{K_1 x_1 + K_2 (1 - x_1)}$$



Temp	Kpropane	khexane	X	Ypropane
0	1.5	0.026	0	0
10	2	0.045	0.1	0.831601
20	2.5	0.07	0.2	0.899281
30	3.2	0.11	0.3	0.925747
40	4	0.15	0.4	0.946746
50	5	0.22	0.5	0.957854
60	5.5	0.28	0.6	0.967175
70	6.5	0.38	0.7	0.975557
80	7.2	0.49	0.8	0.983271
90	8	0.63	0.9	0.991326
100	9.2	0.8	1	1





b. $F = V + L \rightarrow L = F - V$

$$ZF = Vy + LX$$

$$y = \frac{F}{V}z - \frac{L}{V}x$$

$$y = \frac{1}{0.4}z - \frac{\frac{F}{V} - \frac{V}{F}}{\frac{V}{F}}x$$

$$y = \frac{1}{0.4}(0.3) - \frac{1-0.4}{0.4}x$$

$$y = 0.75 - 1.5x \quad \star \text{operating}$$

c. $y_{\text{prop}} = 0.8$

$$z = 0.6$$

$$x = 0.096$$

$$Fz = Vy + (F-V)x$$

$$Fz = Vy + Fx - Vx$$

$$Fz - Fx = Vy - Vx$$

$$\frac{V}{F} = \frac{z-x}{y-x} = \frac{0.6-0.096}{0.8-0.096} = 0.72$$

★ use x value to find temp @ $x = 0.077$
 $T = 7.7^\circ\text{C}$

★ plot operating and find intersection within the x_{prop} vs y_{prop} graph

$$\boxed{x = 0.077}$$

$$\boxed{y = 0.64}$$

d. $\sum \frac{z_i (K_i - 1)}{1 + (K_i - 1)V/F} = 0$

$K_{\text{prop}} \quad 7.7^\circ\text{C} \quad y = 1.5 + \frac{(2-1.5)}{10-0} (7.7-0) = 1.885$

$K_{\text{hex}} \quad 7.7^\circ\text{C} \quad y = 0.026 + \frac{(1.045-0.026)}{10-0} (7.7-0) = 0.0406$

RR B: $\frac{0.3(1.885-1)}{1+0.4(1.885-1)} + \frac{(1-0.3)(0.0406-1)}{1+0.4(0.0406-1)} = -0.8899$

RR C: $\frac{0.6(1.885-1)}{1+0.72(1.885-1)} + \frac{(1-0.6)(0.0406-1)}{1+0.72(0.0406-1)} = -0.901$

Overall spreadsheet

kPa 300

temp	Kpropane	khexane	X	Ypropane	x value
0	1.5	0.026	0	0	0.75 0.077
10	2	0.045	0.1	0.831601	0.6 0.077
20	2.5	0.07	0.2	0.899281	0.45 0.077
30	3.2	0.11	0.3	0.925747	0.3 0.077
40	4	0.15	0.4	0.946746	0.15 0.077
50	5	0.22	0.5	0.957854	0 0.077
60	5.5	0.28	0.6	0.967175	-0.15 0.077
70	6.5	0.38	0.7	0.975557	-0.3 0.077
80	7.2	0.49	0.8	0.983271	-0.45 0.077
90	8	0.63	0.9	0.991326	-0.6 0.077
100	9.2	0.8	1	1	-0.75 0.077

